

AI Audit Report

Faculty of Information Technology (FIT) — Ho Chi Minh City University of Science (HCMUS)
CS423 / CSC13003 — Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — *AI Use Policy Templates for Higher Education*. CC BY-NC-SA 4.0.

This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

1. Student Information

Field	Value
Student name (printed)	Ngô Hồng Thanh
Student ID	23127475
Class / Cohort	CSC13003B
Assignment ID (e.g. HW#00)	HW#01
Assignment date	6/8/2026
AI tool(s) used	ChatGPT, Gemini, Claude

- **AI tool used?** [X] Yes [] No

2. Instructions (read before filling)

- Add **one row per AI-generated artifact** (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the **verbatim prompt** — DO NOT paraphrase.
- Paste the **verbatim AI output** (or include a labelled screenshot in the report).
- Tag the verdict: **VALID / INVALID / INCOMPLETE**.
- Reasoning must cite a **course slide, ISTQB section, or technical RFC**.
- Show the **corrected artifact** with the change highlighted.

Sample row is in *italic* — replace it before submission.

3. Audit Table

#	(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
#0	Tool: Gemini Time: 12:27 AM 07/06/2026 Prompt: "Requirement 1 – QA/QC Job Market 2026+ (40 pts): Find	10 job listing summaries (Job 1–10) with company names, descriptions, required skills, and links (jobs_gemini1.png -> jobs_gemini5.png).	INVALID	Per ISTQB FL §2.1 (Test Planning), artifacts must be traceable to authoritative, live sources. All 10 job links returned "not found" / 404 errors when accessed, the AI failed	I manually searched LinkedIn and company career pages to locate and verify 10 actual active QA/QC job postings published

#	(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	10 QA/QC job postings PUBLISHED WITHIN 60 DAYS... tôi có yêu cầu như trên, tìm và đưa link cho tôi"			to verify link availability before presenting them. This is a fundamental usability failure : the output was unusable as-is and required complete manual replacement. ISTQB FL §4.4.2 (Error Guessing) emphasizes that AI-generated content requires validation; here, zero links survived validation.	within 60 days. Screenshots were retaken with my account name visible in the corner. All 10 postings met the ≥3 AI/LLM-skill requirement. The original AI-generated links were discarded entirely.
#1a	Tool: Gemini Time: 1:30 AM 07/06/2026 Prompt: "với yêu cầu: Requirement 2 – 20 Software Defects 2022–2026 (20 pts): Find 20 software defects... NEW: find 1 place where the AI is biased or hallucinates... Clarification: this applies to EVERY defect... tôi chưa hiểu lắm, giải thích yêu cầu lại cho tôi, ý đề kêu tìm lỗi hả, nhưng tìm lỗi của phần mềm gì?"	2 screenshots: prompt_explain_req2(1).png, prompt_explain_req2(2).png, Gemini clarified the requirement correctly: "AI bias/hallucination" refers to the AI <i>explaining</i> each defect making up (hallucinating) false details about those software defects, not finding bugs in AI itself.	VALID	This was a clarification prompt, not a content-generation task. The student's question reflected genuine confusion about the NEW requirement in HW01_Requirement.md. Gemini answered accurately by distinguishing between: (1) the software defects themselves (the 20 incidents), and (2) the AI's own bias/hallucination when <i>explaining</i> those defects. This is a correct interpretation per ISTQB FL §4.4.2, the student was asking for guidance on how to identify AI-generated errors, which is exactly the skill being assessed.	No fix required, prompt was for clarification only; output accepted as-is.
#1	Tool: Gemini Time: 16:00 07/06/2026 Prompt: "tôi có 20 link về Software Defects 2022–	20 defect explanations (defect1_aircanada_chatbot.png -> defect20_microsoft.png) covering AI/LLM defects (1–5) and general system defects (6–20).	INCOMPLETE	Per ISTQB FL §4.4.2 (Error Guessing), AI-generated content must be validated against authoritative sources. AI hallucinated specific details not present in	Each defect entry in §Requirement 2 was corrected with an AI Bias/Hallucination in explanation column appended,

#	(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
	2026 như sau, hãy giải thích từng cái cho tôi và nguyên nhân, hậu quả, giải pháp trong từng bài: [20 URLs]"			source articles (e.g., Defect 1: fabricated passenger name "Jake Moffatt" and "buy-then-refund" instruction; Defect 2: fabricated transfer_funds() function; Defect 9: mischaracterized supply chain attack as credential compromise instead of GitHub Actions + cache poisoning; Defect 10: conflated Confluence vulnerability with Questions-for-Confluence hardcoded password). ISTQB FL §2.1 requires testers to verify traceability between test evidence and requirements, a principle that extends to verifying AI output against primary sources.	citing which specific details were fabricated or overstated vs. what the source article actually states. Example fix for Defect 1: removed fabricated "buy-then-refund" instruction; noted that the article only states chatbot offered a post-flight refund claim, not a purchase-first workflow.
#2	Tool: ChatGPT Time: 12:38 AM 08/06/2026 Prompt: "viết cho tôi 15 test case cho 1 chiếc máy quạt phổ thông, bao gồm đủ các thông tin (Objective / Input / Steps / Expected / Actual / Verdict). Máy quạt không có remote, bật bằng nút vặn 4 mức 0 (tắt), 1, 2, 3. có nút nhấn	15 test cases with columns: Objective / Input / Steps / Expected / Actual / Verdict (screenshots: ai_screenshot_testcases-missing1.png, ai_screenshot_testcases-missing2.png).	INCOMPLETE	ISTQB FL §4.3 (Boundary Value Analysis) and §4.4.1 (Equivalence Partitioning) require coverage of edge and corner cases. AI generated 15 TCs covering nominal paths but missed ≥ 3 edge cases: (a) TC10 – oscillation switch activated while fan is OFF (unpowered motor); (b) TC15 – tilt angle beyond safe mechanical limit causing tip-over; (c) TC16 – power interruption mid-operation followed by restore. Per ISTQB FL	Added TC10: Oscillation ON while fan speed = 0 (OFF) → Expected: oscillation motor does not activate. Added TC15: Tilt head to maximum tilt angle → Expected: fan should not tip over; verify mechanical stop works. Added TC16: Simulate sudden power loss during operation → Expected: fan stops safely with no unexpected restart. Also renamed all Objective fields to

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	xuống/kéo lên để bật tắt chế độ quay trái phải. có thể chỉnh đầu quạt gập lên, xuống, có thể điều chỉnh độ cao của quạt."			§2.2, test conditions must include both valid and invalid partitions AI only explored the valid partition.	assertive statements (e.g., "Verify fan oscillates when oscillation button is pressed at speed ≥ 1" instead of "Testing oscillation").
#3	Tool: Claude Time: 4:08 AM 08/06/2026 Prompt: "tạo 1 file md và vẽ cho tôi 1 ISTQB QA/QC role mindmap. đặt file ở HW01-Software-Testing"	ISTQB QA_QC Role mindmap.md — a hierarchical mindmap categorizing ISTQB roles, QA vs QC responsibilities, verification/validation focus areas, and CTAL tracks.	INCOMPLETE	ISTQB Foundation Level Syllabus explicitly defines only two role clusters: Test Management Role and Testing Role (FL Syllabus §2.2). "QA" and "QC" are not ISTQB role names — they are organizational umbrella terms. This is a fundamental terminology error. Additionally, ISTQB CTAL Test Analyst syllabus does not make "Test Process Improvement" a primary focus, and UAT belongs under Validation (V-Model), not Verification — contradicting ISTQB FL §1.3 (Verification vs. Validation distinction). AI conflated industry colloquialisms with ISTQB taxonomy.	Replaced the mindmap with a corrected version using only ISTQB-accurate role terminology: Test Manager (Test Management Role) and Test Analyst / Test Automation Engineer / Test Consultant (Testing Role). Moved UAT from Verification to Validation. Removed Test Process Improvement from CTAL Test Analyst track and relocated it to Test Manager track. Verified against ISTQB FL Syllabus §2.2 and CTAL syllabus.
#4	Tool: ChatGPT Time: 4:32 AM 08/06/2026 Prompt: "tìm 3 mistake trong ISTQB QA/QC Role Mindmap sau: [mermaid code]"	3 screenshots (find_mistake1.png → find_mistake3.png) identifying Mistake #1 (QA/QC not ISTQB Roles), Mistake #2 (UAT under Verification instead of Validation), and Mistake #3 (Test Process Improvement misplaced in CTAL Test Analyst Track).	VALID	All 3 mistakes were correctly identified and cited against ISTQB FL Syllabus §2.2 (role taxonomy), §1.3 (Verification vs. Validation), and CTAL Test Analyst syllabus. The analysis was thorough and aligned with ISTQB standards — no fabricated details, no	No fix required; output accepted as-is.

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				overstatements. Output was accepted as-is.	

4. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	6	100 %
VALID (correct, accepted as-is)	2	33 %
INVALID (wrong; rejected)	1	17 %
INCOMPLETE (acceptable after edits)	3	50 %

Note: 1 artifact was tagged **VALID** (accepted as-is; clarification prompt answered correctly). 1 artifact was tagged **INVALID** (rejected entirely; manually re-done). 3 artifacts were tagged **INCOMPLETE** (acceptable after student corrections).

5. Conclusion — When should AI be used (or not)?

AI proved highly effective as a **brainstorming accelerator** for generating initial content (defect explanations, test case templates, structural outlines), allowing students to reach a first draft quickly. However, every artifact audited contained at least one factual error or structural gap that required manual correction. The most persistent failure mode was **AI hallucination** — fabricating specific details (names, dates, code snippets, technical mechanisms) not present in source articles — combined with **terminology imprecision** (e.g., using "QA/QC" instead of ISTQB role taxonomy). For job postings, AI produced a complete failure: all 10 links returned 404, making the output entirely unusable (tagged **INVALID**). AI also systematically missed **edge/corner cases** when generating test cases, focusing on nominal paths and valid partitions while ignoring invalid ones. **Recommendation:** Use AI for ideation and scaffolding, but always validate output against primary sources and apply ISTQB techniques (boundary value analysis, equivalence partitioning, error guessing) manually to cover gaps. AI should never replace the tester's critical judgment or traceability verification step. For tasks requiring live web data (job postings), AI is unreliable and manual search is preferred.

6. Mandatory Disclosure (*paste verbatim*)

"Report sections §Requirement 2 (20 Software Defects), §Requirement 3 (Test Cases for Physical Product), §ISTQB QA/QC Role Mindmap, and §Mistake Analysis (Mindmap Review) were initially generated by AI tools (Gemini, ChatGPT, Claude); I reviewed and corrected each artifact. For §Requirement 2, I appended an 'AI Bias/Hallucination in explanation' column to each defect entry, identifying fabricated or overstated details against primary source articles. For §Requirement 3, I added 3 missing edge cases (TC10, TC15, TC16) and renamed all Test Case Objectives to assertive statements. For the Mindmap, I identified 3 factual mistakes per ISTQB FL Syllabus §2.2 and CTAL syllabus and replaced the mindmap with a corrected version. For §Mistake Analysis, ChatGPT correctly identified all 3 mistakes and was accepted as-is with no corrections needed. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category."

Signature

Field	Value
Student name (printed)	Ngô Hồng Thanh
Student ID	23127475
Class / Cohort	CSC13003B
Course	CS423 / CSC13003 – Software Testing
Instructor	MSc. Trần Thị Bích Hạnh, MSc. Hồ Tuấn Thanh
Date	6/8/2026
Signature	Ngô Hồng Thanh

References

- Kharbach, M. (2026). *AI Use Policy Templates for Higher Education*. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). *OECD Education Working Paper No. 338*.
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- Anthropic (2025). *Building reliable AI test agents* — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.